

Yejun (Henry) Kim

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Education

University of Virginia, Charlottesville, VA 2021–2025

Bachelor of Science, Aerospace Engineering & Mathematics

Cumulative GPA: 3.6 / 4.0

Honors: Dean's List, QuestBridge National College Match Scholar

Relevant Coursework: Thermodynamics, Computer Aided Design, Statics, Dynamics, Strength of Materials, Aerospace Structures, Aerospace Materials, Aerodynamics, Thermal Fluids, Flight Vehicle Dynamics

Work Experience

Epic Systems | Technical Solutions Engineer Madison, WI | Jul 2025–Present

- Resolved high-severity production issues in medical scheduling software via backend M programming, data validation, and root-cause analysis across 200+ databases

We-Xite Lab | Undergraduate Researcher Storrs, CT | Summer 2024

- Fabricated a vibration isolation system and applied 2D FFT analysis in MATLAB to characterize frequency response
- Built and validated an orientation control system for a submersible turbine

Factory Mutual Insurance | Reliability Engineering Intern Norwood, MA | Summer 2023

- Modeled component lifetimes using Weibull, exponential, and lognormal distributions; automated reliability calculations and statistical transformations using VBA

Projects & Extracurriculars

AquaMetric Automatic Water Dispensing System Fall 2024

- Derived a Bernoulli/continuity-based theoretical model relating tank height, volume, and drain time for a trapezoidal frustum geometry; corrected the model with multivariate polynomial regression (cubic for volume, quintic for time) in Python/scikit-learn
- Reduced time-prediction error from an MSE of 10.2 (uncorrected theoretical model) to 0.297 (regression-corrected), achieving 19 mL dispensing accuracy validated across 60+ trial measurements at varying heights
- Designed and 3D-printed a circuit enclosure in SolidWorks housing a pressure sensor and servo-actuated spigot; implemented Secant-method root-finding in C++ to compute required drain time from user-input volume

RASC-AL Capstone Project | Electrical and Controls Lead 2024-2025

- Led the Electrostatic Precipitator, Software/Controls, and Power subteam (3-person team) for MARRS, a Mars dust-mitigation concept submitted to NASA's RASC-AL competition
- Performed structural analysis (buckling, bending stress, deflection) on an aluminum support frame under Martian wind and payload loads, validating a 0.7–1.1 mm deflection margin against material yield limits
- Derived operating voltage requirements for a parallel-plate electrostatic precipitator from particle kinematics and Paschen's Law, establishing a 1,030–8,910 V design window validated against simulated Martian atmospheric conditions (610 Pa, CO₂)

Skills

Tools: Simulink, SolidWorks, ANSYS, Raspberry Pi, KiCad

Languages: MATLAB, R, C++, Python, SQL

Frameworks/Libraries: NumPy, pandas, scikit-learn, OpenCV